

Food Nutrition & Health Risk Analytics System

Analyzing nutritional quality and health risks using data analytics

PYTHON

PANDAS

MYSQL

POWER BI

GITHUB

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Problem, Objectives & Dataset at a Glance

Problem

Consumers lack clear, data-driven insights into the nutritional quality and health risks of packaged and traditional foods.

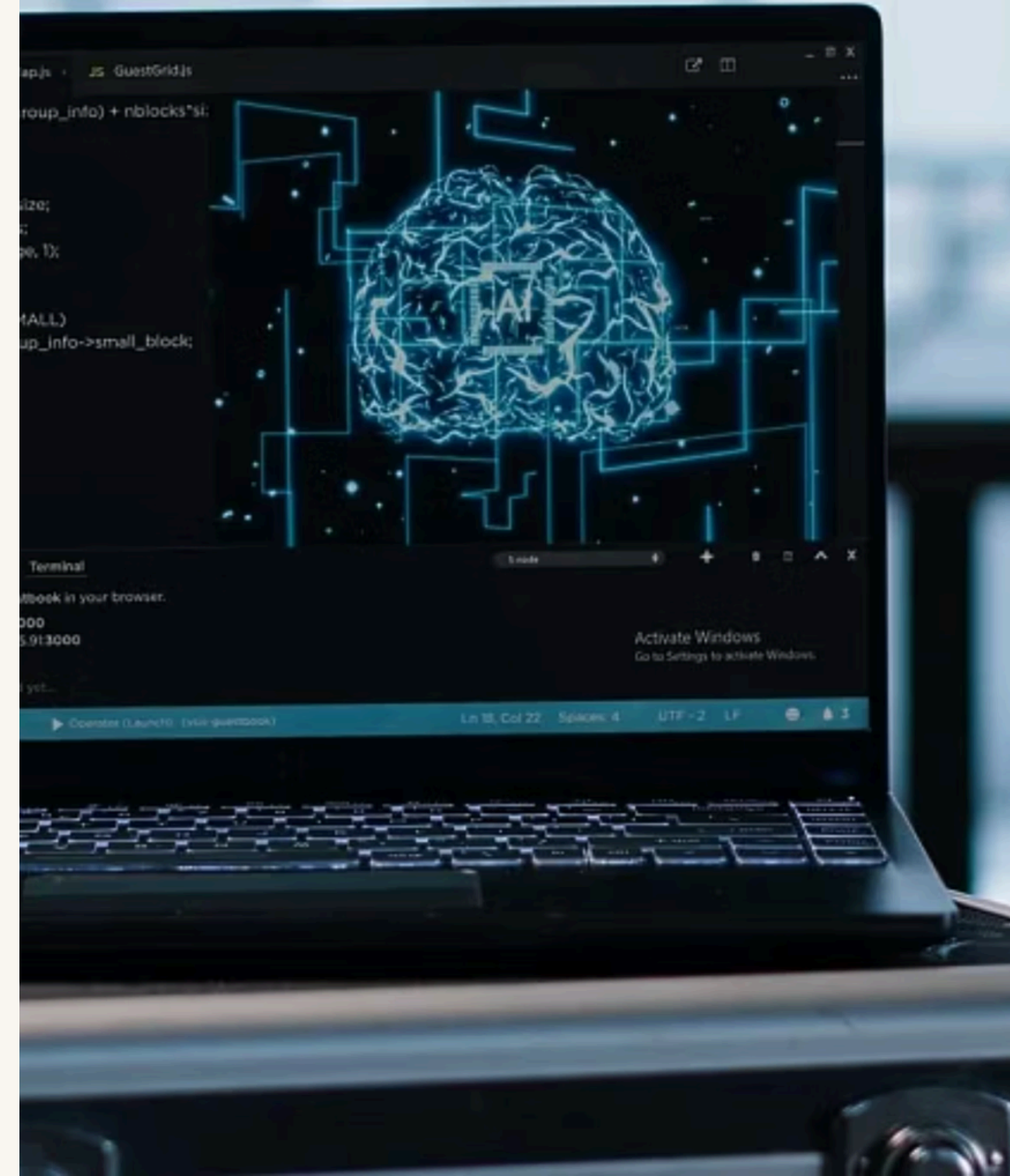
Project Objectives

- Analyze 1,028 food items across 9 nutrients
- Develop Health Risk Score & Nutrition Density
- Identify healthy vs high-risk foods
- Deliver interactive Power BI dashboards

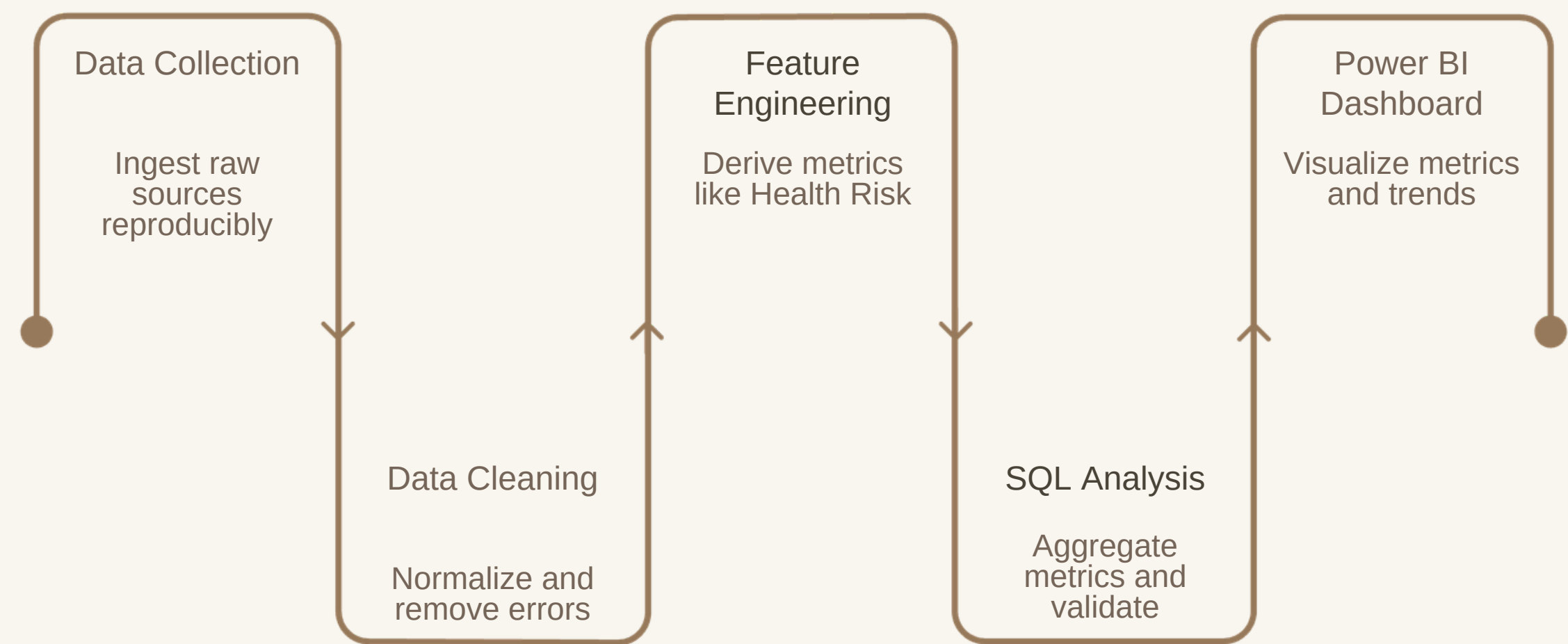
Dataset Overview

1,028 items · 9 core nutritional features · no missing values

Key features: Energy, Protein, Fat, Carbs, Free Sugar, Fibre, Cholesterol, Calcium



Project Workflow & Feature Engineering



The pipeline enforces reproducible ETL, metric derivation, and dashboard-driven exploration. Key engineered metrics are shown below.



Health Risk Score

Composite score combining normalized Fat, Free Sugar, and Cholesterol with domain-weighted coefficients to flag high-risk items.

Nutrition Density

Ratio of beneficial nutrients (Protein, Fibre, Calcium) to energy; rewards nutrient-rich, lower-energy foods.

Validation

Checks include distribution analysis, correlation matrices, and SQL-based consistency tests.

Top Nutrient-Dense Foods

Ranked analysis highlights traditional pulse- and grain-based preparations as nutrient-dense leaders.



Maa Chaane Ki Dal

Top nutrition density: high protein and fibre relative to energy.



Cracked Wheat & Green Gram Premix

Balanced macro profile with excellent micronutrient density.



Dessicated Coconut Chutney

Surprisingly nutrient-dense when portioned appropriately.

Foods with Highest Health Risk Scores

Highest risk items are dominated by high free sugar and saturated fat content.



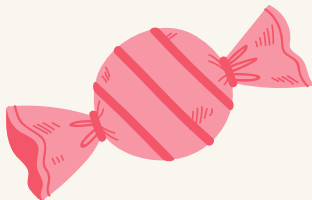
Butter Icing

Extremely high sugar + saturated fat → top health risk.



Glaze Icing

High free sugar concentration, low nutrient value.



Mango Murabba Candy

Sugar-preserved fruit with concentrated free sugars and energy.

Healthy Food Recommendations

High Protein · Low Risk

Roast Chicken

Lean protein, low added sugars — strong nutrition / low risk.

Tandoori Chicken

Spice-forward preparation with preserved protein profile.

Fish Tikka

High protein, beneficial fats, low processed sugar.

High Fibre · Low Sugar

Maa Chaane Ki Dal

Top fibre and protein—consistent low-risk performer.

Cracked Wheat & Green Gram

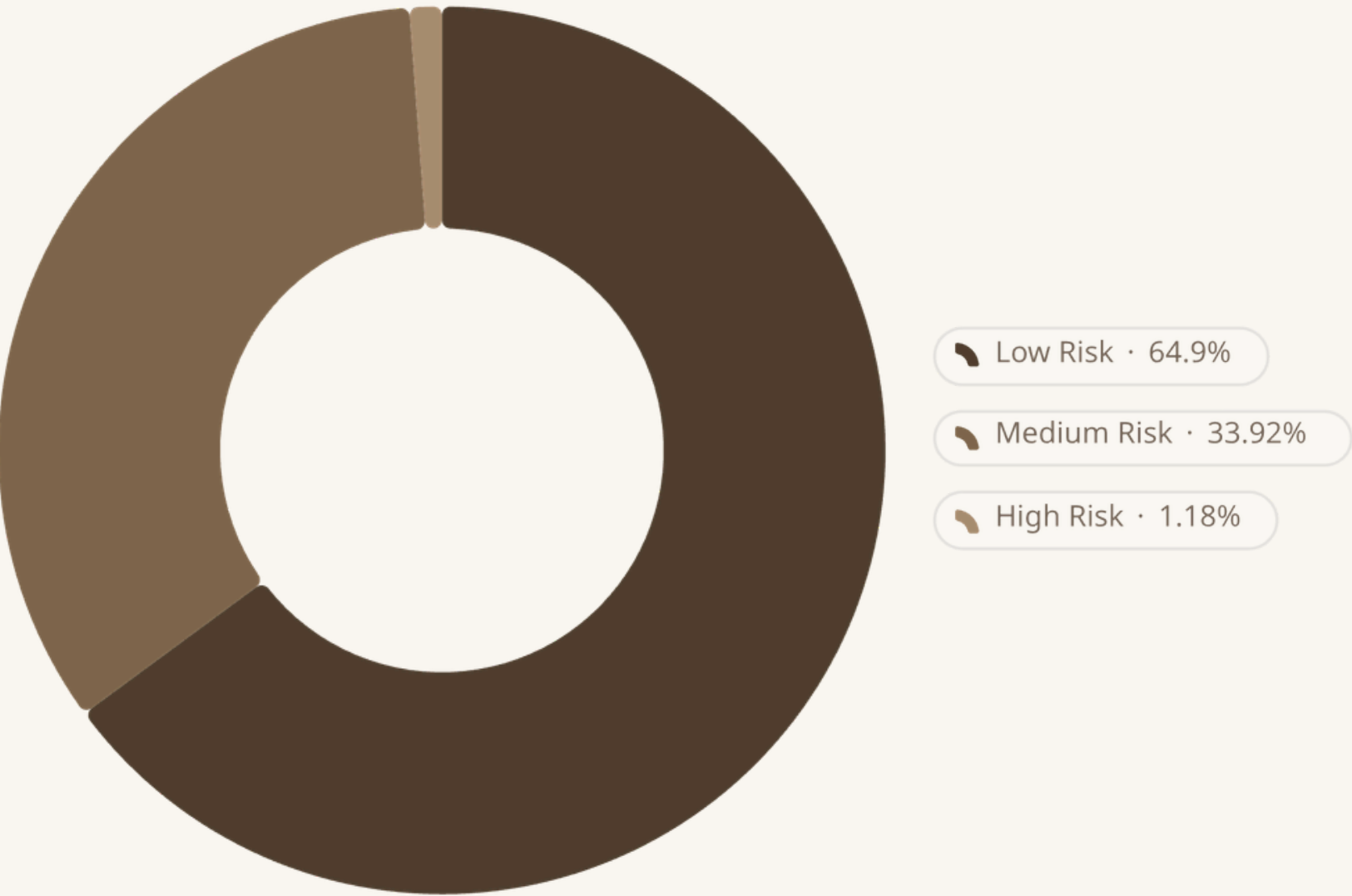
Whole-grain base with sustained energy release and fibre.

Soya Roti

Plant protein with high fibre and micronutrient content.

Recommendation: Prioritise whole pulses, lean proteins, and fibre-rich grains for lower health risk and improved nutrition density.

Health Risk Distribution & Leaderboard



Distribution shows the majority of items classify as Low Risk.

Top Healthiest Foods



1. Maa Chaane Ki Dal

Highest Nutrition Density & low Health Risk.



2. Cracked Wheat & Green Gram Premix

Balanced macros and fibre-forward profile.



3. Dessicated Coconut Chutney

High nutrient concentration in controlled portions.

Power BI Dashboard — Interactive Showcase



KPI Cards

Key metrics: % Low Risk foods, Avg Nutrition Density, Avg Health Risk Score — at-a-glance monitoring.

Interactive Filters

Filter by category, region, serving size, and nutrient thresholds for targeted analysis.

Visuals

Risk distribution, ranked leaderboards, and nutrient correlation panels support rapid insights.

Conclusions & Next Steps

01

Key Conclusions

Traditional pulse- and grain-based foods scored highest on Nutrition Density; high-sugar/fat items had the highest Health Risk Scores. 662 of 1,028 foods classified as Low Risk.

02

Future Enhancements

Apply machine learning for risk prediction, build personalised recommendations, integrate real-time nutrition APIs, and deploy a web-based interactive dashboard.